

Name: \_\_\_\_\_

Roll number: \_\_\_\_\_

Time Allowed: 15 min

Total Marks: 20

**Question#1 (3+3+4=10 Marks)**

----- CLO3

Write down the output of the following codes:

|                                                                                                                                                                                                                                                                                                                                                                    |                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <pre>int main() {     int arr[5]={4,6,5,1,9};     int *p *q;    int a ,b;     p= arr+1;     q=arr+3;     cout &lt;&lt; *p &lt;&lt; " " &lt;&lt; *q &lt;&lt; endl;     a=*(p+3);    b=(*q)+3;     cout &lt;&lt; a &lt;&lt; " " &lt;&lt; b &lt;&lt; endl;     p=p+2;      q=q-*q;     cout &lt;&lt; *p &lt;&lt; " " &lt;&lt; *q &lt;&lt; endl;     return 0; }</pre> | <pre>6 1 9 4 1 5</pre>           |
| <pre>int main(){     int x = 10, y = 20, z = 30;     int * ptr;     cout &lt;&lt; x &lt;&lt; " " &lt;&lt; y &lt;&lt; " " &lt;&lt; z &lt;&lt; endl;     ptr = &amp;x;     *ptr *= 10;     ptr = &amp;y;     *ptr += 5;     ptr = &amp;z;     *ptr *= 2;     cout &lt;&lt; x &lt;&lt; " " &lt;&lt; y &lt;&lt; " " &lt;&lt; z &lt;&lt; endl;     return 0; }</pre>    | <pre>10  20  30 100 25  60</pre> |
| <pre>void inc_count(int *count_ptr) {     (*count_ptr)++; } int main() {     int count = 0;     while (count &lt; 5)         inc_count(&amp;count);     cout &lt;&lt; count &lt;&lt; endl;     return 0; }</pre>                                                                                                                                                   | <pre>5</pre>                     |

**Question#2 (10 Marks)**

----- CLO3

Considering the following code:

|                                                                                       |                                                                                   |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <pre>struct Item {     char name[50];     int quantity;     float unitPrice; };</pre> | <pre>struct Cart {     Item * data;     int noOfItems;     int capacity; };</pre> |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|

Write down the definition of the following function, it will Increase the capacity of the Cart by s (default value is 5). Ensure that the existing data is preserved correctly.

**void resize(Cart & c, int s=5);**

```
void resize(Cart &c, int s = 5)
{
    // New capacity
    int newCapacity = c.capacity + s;

    // Allocate new array with increased capacity
    Item* newData = new Item[newCapacity];

    // Copy existing items into new array
    for (int i = 0; i < c.noOfItems; i++)
    {
        newData[i] = c.data[i];
    }

    // Deallocate old memory
    delete[] c.data;

    // Update cart to point to new array
    c.data = newData;
    c.capacity = newCapacity;
}
```